Worldline

UNDER THE POLE
INNOVATIVE UNDERWATER EXPLORATION



Onboarding with the Explorers

EXPLORE

CSR by Worldline

The Explorers of Under The Pole Expedition

A family of adventurers driven by a genuine pioneering spirit !



Emmanuelle PÉRIÉ-BARDOUT, 38 years old
Co-Initiator and Co-Director of Under The Pole expeditions
A skipper with a degree from the French National School of Sailing and a specialist in polar regions
& Robin BARDOUT

Ghislain BARDOUT, 37 years old
Initiator and Director of Under The Pole expeditions
A graduate of the Lausanne Polytechnic School in energy engineering, a specialist in deep polar diving and an undersea cameraman
& Tom BARDOUT

Kayak BARDOUT

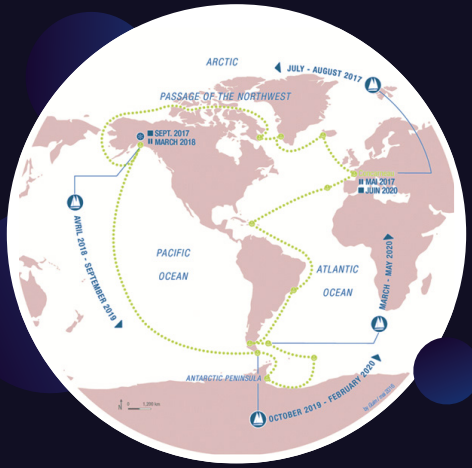


Meet our Team of Permanent Explorers

Karyn ROGER - Administration
Sylvain PUJOLLE - Technics & Engineer
Erwan MARIVINT - Partners & Logistics
Cécile CLIQUET - Sponsoring
Victor RAULT - Films & Communication
Franck GAZZOLA - Photography & Intern. Develop.
Jérémy FAUCHET - Science & Pedagogy
Kayak BARDOUT - Expedition Mascot !



Follow our eXpedition #1: The Journey



From 2017 to 2020, **Under The Pole embarks on an extraordinary adventure** dedicated to
exploring our oceans.

Our team of divers and scientists will sail around the world for three years on board the WHY polar schooner, from the Arctic to the Antarctic, by way of the Pacific and Atlantic Oceans.

Our goal: explore the ecosystems of the Twilight Zone at all latitudes.

Follow our eXpedition #2: The Key Figures

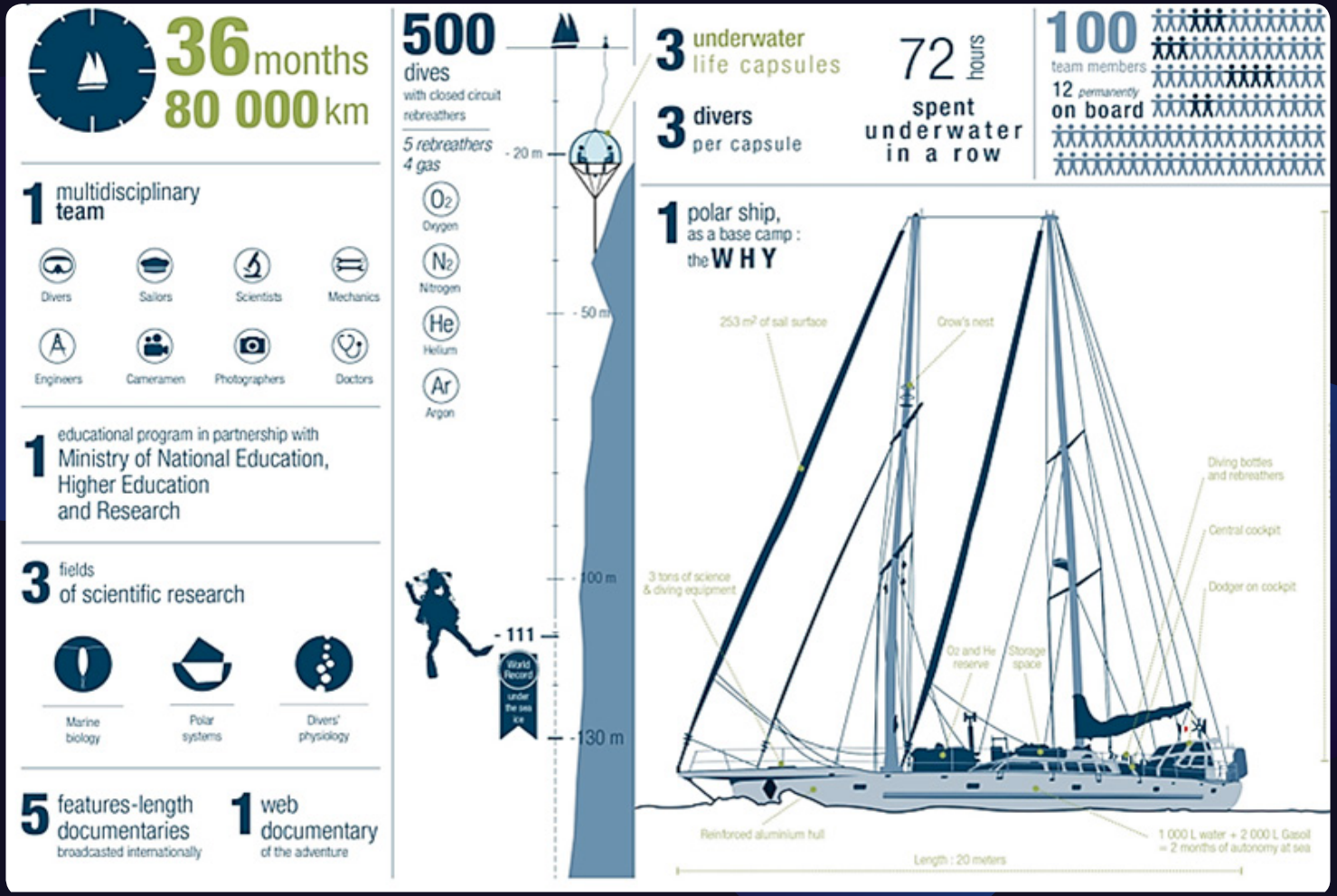


THE "WHY"

The **WHY** is the expedition ship, a robust aluminum schooner designed to navigate in seas around the world, from the tropics to the polar regions, or to winter in the ice caps.

A veritable itinerant base camp, the ship can hold **12 crew members** in five cabins and carry several tons of material dedicated to undersea and land exploration, the implementation of scientific programs and the creation of films and reports.

The ship is equipped as a pleasure craft for training in sailing and diving in the **"A" construction category.**


36 months
80 000 km
1 multidisciplinary team
Divers
Sailors
Scientists
Mechanics
Engineers
Cameramen
Photographers
Doctors
1 educational program in partnership with Ministry of National Education, Higher Education and Research
3 fields of scientific research
Marine biology
Polar systems
Divers' physiology
5 features-length documentaries broadcasted internationally
1 web documentary of the adventure
500 dives with closed circuit rebreathers
5 rebreathers
4 gas
O2 Oxygen
N2 Nitrogen
He Helium
Ar Argon
- 20 m
- 50 m
- 100 m
- 111
World Record under the sea ice
- 130 m
3 underwater life capsules
3 divers per capsule
72 hours spent underwater in a row
100 team members
12 permanently on board
1 polar ship, as a base camp : the WHY
253 m² of sail surface
Crow's nest
Diving bottles and rebreathers
Central cockpit
Dodger on cockpit
3 tons of science & diving equipment
O2 and He reserve
Storage space
Reinforced aluminium hull
1 000 L water + 2 000 L Gasoil = 2 months of autonomy at sea
Length : 20 meters

Under The Pole is supported by Explore

Created in 2013 by **Roland Jourdain,**
Explore supports the expedition « Under The Pole ».



Explore organization aims to :

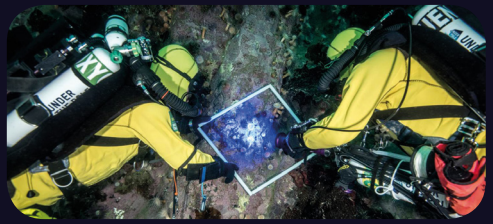
- Develop Research, Innovations and Sensitization actions in connection with environmental stakes,
- Develop, select and support environmental projects proposed by international stakeholders across the world,
- ... and create an Explorers community for the Human beings and the planet.

EXPLORE





Discover our 3-year missions



Research & Development

- Invent, test and use operational, innovative tools dedicated to the undersea exploration of tomorrow.
- Share techniques and validated protocols to accelerate the acquisition of new knowledge.

Science

- Apply our technical expertise to scientific research in the domains of oceanography, polar systems and hyperbaric physiology.
- Better understand our oceans and their role in the climate balance in order to face ecological and societal challenges.





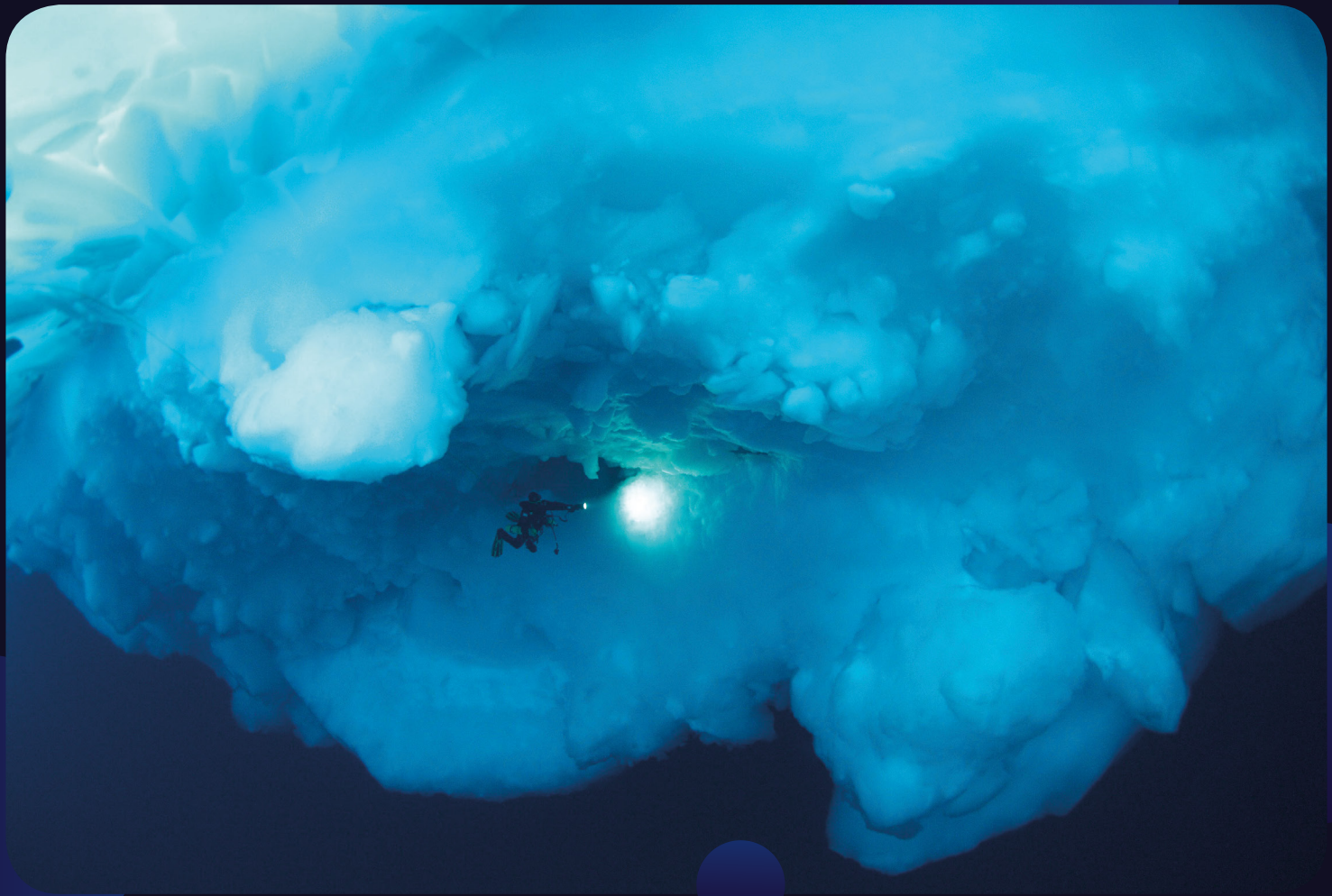
Documentary

- Make a film and photographic opus that illustrates the story of discoveries and the expedition highlights.
- Distribute it internationally to all audiences for a high-profile media impact.

Education

- Stimulate and inspire younger generations, and transfer our passion to them.
- Keep a record and create tools to make audiences aware of climate change and the actions needed to halt it.







WHAT IS THE TWILIGHT ZONE ?

Share our discoveries in the Twilight Zone

- **90% of the ocean** remains to be explored, while it is an incredible resource: climate regulator, carbon sinks, fish resources that are the main source of animal proteins for a billion people ... Explore the twilight zone is necessary to learn more about this resource that is essential to us, to discover new species and ecosystems and to better observe the extent of the changes that it undergoes.

- **From 2017 to 2020,** scientists who are part of the journey will study biofluorescence, large sharks, and deep corals. The results of the research will be accessible to the scientific community through the publications of the scientific explorers and to the general public through images and documentaries





50m

150m

Our Scientist Program in the Arctic Ocean

- **Bioluminescence and Natural Biofluorescence in the Arctic:** discover and study for the first time biofluorescent and bioluminescent species in polar regions.

  Understanding the functioning of biofluorescence in animals allows us to translate this knowledge into the medical and biological field. In the laboratory, fluorescence can be used to monitor the evolution of a virus or bacterium during the infection of an animal, thus contributing to the development of antivirals or antibiotics. It can also be used in the monitoring of cancer cells and in the development of new treatments.



- **Inventory of fauna and flora of Arctic marine species:** study of the bathymetric distri bution of species on different habitats and search for new species in the Twilight zone of the polar regions.

  The goal is to inventory the natural resources present in the Arctic marine ecosys- tems, areas that are still very little studied.





Our Scientist Program in the Pacific Ocean

• **Deep Coral Study** (MCE - Mesophotic Coral Ecosystems)

• **Behavioral study of large hammerhead sharks and bulldog sharks,** species having the common characteristics of living in significant depths, thus making it difficult to approach them.

• **“CAPSULE” project:** with the double objective of innovating in diving techniques and studying the physiology of divers, a «capsule» of life at saturation will be tested in warm waters during the Polynesian step point.

└ So far, it is a technique used exclusively in industrial diving on oil or gas platforms but which, applied in a lighter way, could revolutionize the scientific studies of the submarine environment, particularly with significant depths.







Worldline, our IT partner to accelerate innovation

1. Under The Pole is a submarine expedition looking for an **IT partner** to leverage scientist discoveries

2. Under The Pole aims to **promote** the protection of the planet through the **broadcast of media on the internet**

3. Under The Pole has to **track and secure its production of data** and media content

4. Under The Pole has a specific mission to **raise awareness among young public** to protect the nature





Worldline's IT advanced solutions for Polar Discoveries

Leverage scientist discoveries 2018

Data collection server

Centralize in a Worldline server all the data of the expedition collected on PC/MAC on the boat and automatically consolidated in a secured way.

Broadcast of media on the internet

Interactive web map

Internet web page positioning on a map (log book + Google map) all media content of the exploration: pictures, audio, videos, etc.

Track and secure its production of data and media content

Blockchain secured flag

Blockchain solution to be used to create dedicated and single flag for each data provided by the adventure.

Raise awareness among young public

Heasy Robot

Heasy Robot provides a technological window for this adventure and highlights best pictures and video of the nature through an educational and fun media during events at schools.



Explorers visiting Worldline colleagues !



2018 Roadshow dates

9-10th of January / Chantilly / Management Seminar

6 to 8th of March / Sitges / Sales Events

13th of February / Bezons / From the Pole !

27th of June / Brussels / From the Pole !

22nd of June / Frankfurt / From the Pole !

October / International Visio

2019 Roadshow dates

January / APAC / From the Pole !

March / Mumbai / From the Pole !

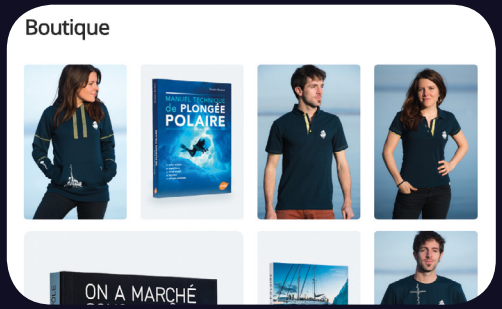
June / Utrecht / From the Pole !

October / Seclin / From the Pole !



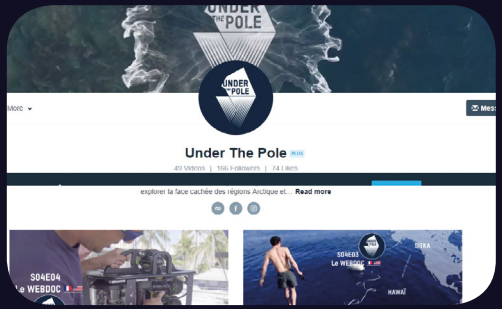
Useful links

Under The Pole Shop: Get your T-Shirt!



shop.underthepole.com

Live the eXperience in Viméo !

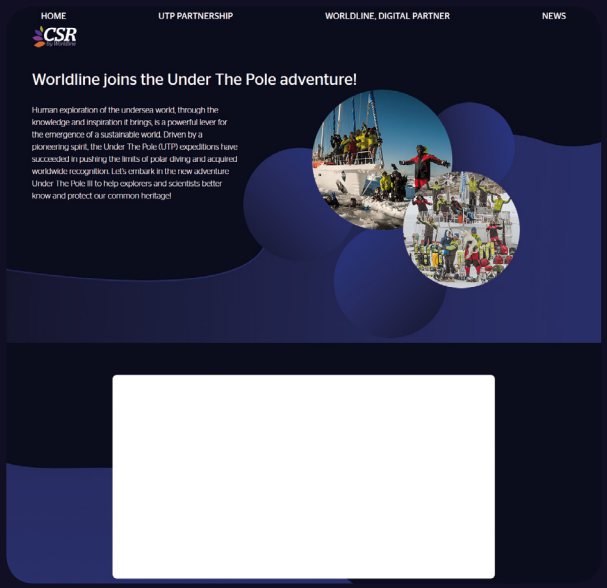


vimeo.com/underthepole

For any further information or suggestion, feel free to contact the Worldline Polar Team **csr@worldline.com**

Stay connected with the Explorers !

Follow us on your dedicated Worldline webpage :
worldline.com/utp



Also accessible via the Intranet
And share our partnership on social networks!

The Worldline Polar Team !

Nicolas Kozakiewicz
Innovation & Development

Bourjaili Cendrella
Expedition communication

Ghislain Bardout



Roland Jourdain

Sébastien Mandron
CSR officer

Pierre-Francois Rudant
Technological partnership

Lise Besné
Expedition coordination

CSR by Worldline

